

RESEARCH REPORT

EU Trade Dependency Analysis

Trade Dependencies and Economic Resilience Across Global Partners
(2015-2024)

Based on UN Comtrade Database

Analyzing trade relationships across France, Germany, Italy, and Spain

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1. Introduction

1.1 Research Context and Motivation

International trade connects modern economies, but this connection creates risks. When one country dominates the supply of critical products like energy or pharmaceuticals, disruptions can cause serious problems. Recent events prove this: Russia's invasion of Ukraine exposed Europe's energy dependence, and COVID-19 revealed weaknesses in electronics and pharmaceutical supply chains.

I chose this topic because trade vulnerabilities directly affect energy prices, medicine availability, and jobs. As an economic intelligence student, I wanted to analyze real data on Europe's weak points rather than just reading news headlines. Brexit gives us enough distance to see how trade patterns shifted.

This report examines how four major EU economies: France, Germany, Italy, and Spain, source ten critical products from global partners between 2015 and 2024. The goal is simple: identify where dependencies are too high and understand how trade patterns shifted after Brexit.

1.2 Research Questions

- **Trade Dependency:** Which EU countries rely too heavily on single partners for critical products? Are supply chains becoming more or less concentrated?
- **Brexit Impact:** How did EU-UK trade change after 2020? Did EU countries successfully replace UK imports with other partners?

1.3 Report Structure

Section 2 describes the data and methods. Section 3 presents the results, first on trade dependencies, then on Brexit impacts. Section 4 discusses findings and policy implications. Section 5 concludes.

2. Methodology

2.1 Data Source and Collection

All trade data comes from the UN Comtrade database, accessed through their advanced bulk download system. The dataset covers bilateral trade flows between four EU countries (France, Germany, Italy, Spain) and ten global partners (USA, China, Russia, UK, Japan, India, Taiwan, Nigeria, South Africa, Egypt) from 2015 to 2024.

2.2 Data Scope and Coverage

The analysis focuses on ten product categories based on HS 2-digit codes: Fruits (08), Cereals (10), Minerals (26), Fuels (27), Pharmaceuticals (30), Plastics (39), Textiles (63), Machinery (84), Electronics (85), and Vehicles (87). Both import and export flows are included.

2.3 Data Cleaning and Preparation

Raw data contained 7,059 records. I filtered out transactions below \$1,000 to focus on meaningful trade relationships, leaving 6,977 records. Product codes were mapped to readable category names. No major data quality issues were encountered.

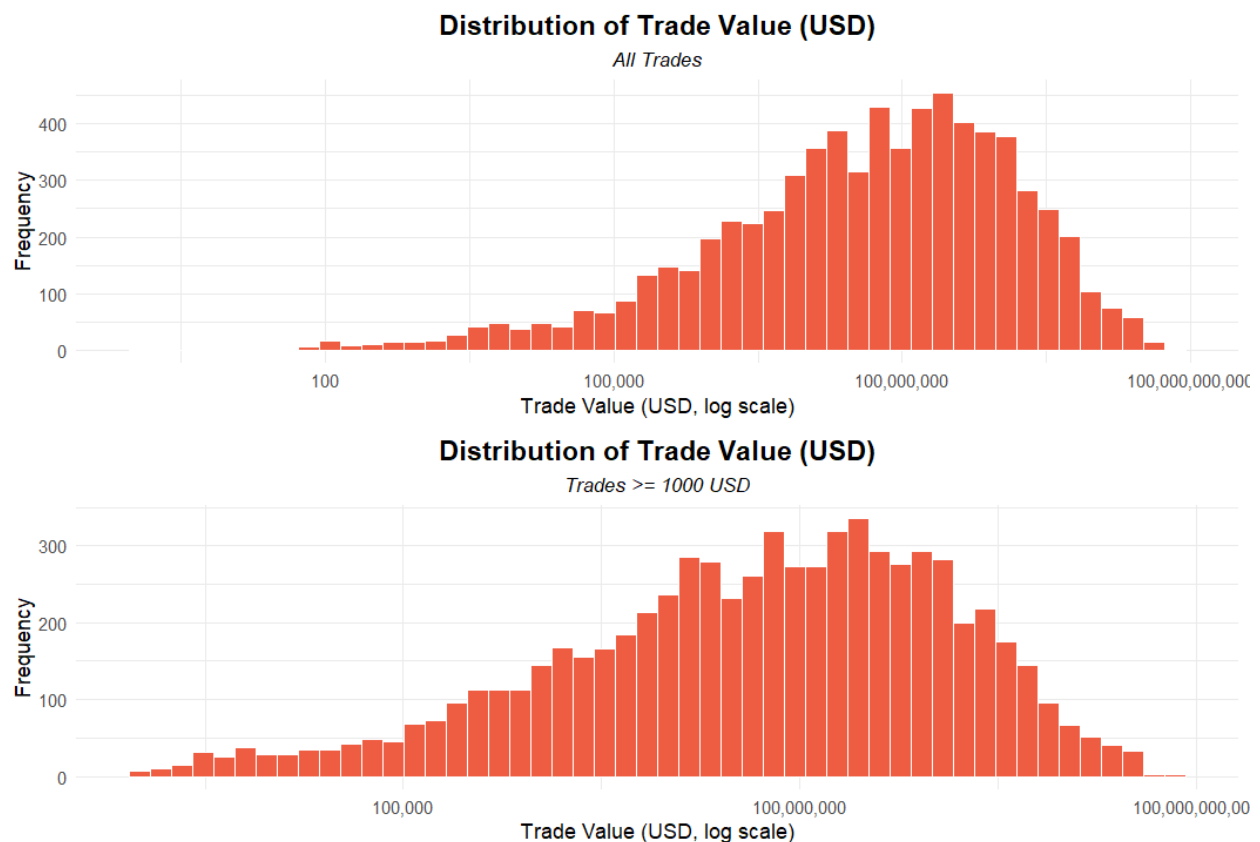


Figure 1: Distribution of Trade Values Before and After Filtering

Limitations:

- **Time period choice:** I used 2023 rather than 2024 for post-Brexit comparisons to ensure complete annual data and avoid reporting delays, while providing more distance from the immediate Brexit chaos.
- **Partner coverage:** The ten partners analyzed are a subset of all trading relationships, major economies like Canada or Brazil were excluded, potentially missing diversification to untracked partners.

- **Multiple events:** Isolating Brexit's impact is impossible since the 2020 drop reflects both Brexit and COVID-19, while 2022-2023 includes Russia-Ukraine war effects.
- **Missing factors:** Trade values don't show supply chain resilience factors like stockpiling or alternative suppliers.

2.4 Analytical Framework

For dependency analysis, I calculated each partner's share of total imports per product-country combination. Shares above 60% indicate concentrated supply chains. For Brexit analysis, I compared 2019 (pre-Brexit) with 2023 (post-Brexit) to measure percentage changes in trade volumes and identify partner substitution patterns. All analysis was conducted in R.

3. Results and Analysis

3.1 Trade Dependency and Concentration

3.1.1 Overall Dependency Patterns

Figure 2 shows the 20 highest import dependencies. China dominates textiles across all four EU countries: Germany (83%), France (83%), Spain (80%), and Italy (79%). Electronics follow the same pattern: Spain (80%), Italy (77%), Germany (72%), and France (70%) all heavily depend on Chinese imports.

The USA dominates different categories. Spain sources 81% of fruits and 79% of pharmaceuticals from the USA. Italy, Germany, and France also rely on the USA for 69%, 69%, and 55% of pharmaceuticals respectively.

Key finding: All four countries show identical vulnerabilities. The same product-partner pairs repeat across countries.

Russia appears only once in the top dependencies with Italy importing 62% of its fuels from Russia. Looking at the underlying data, Germany (52%), France (31%), and Spain (20%) also import fuels from Russia, but at lower shares that don't reach the high-risk threshold. This means Italy has the most concentrated Russian fuel dependency among the four countries analyzed.

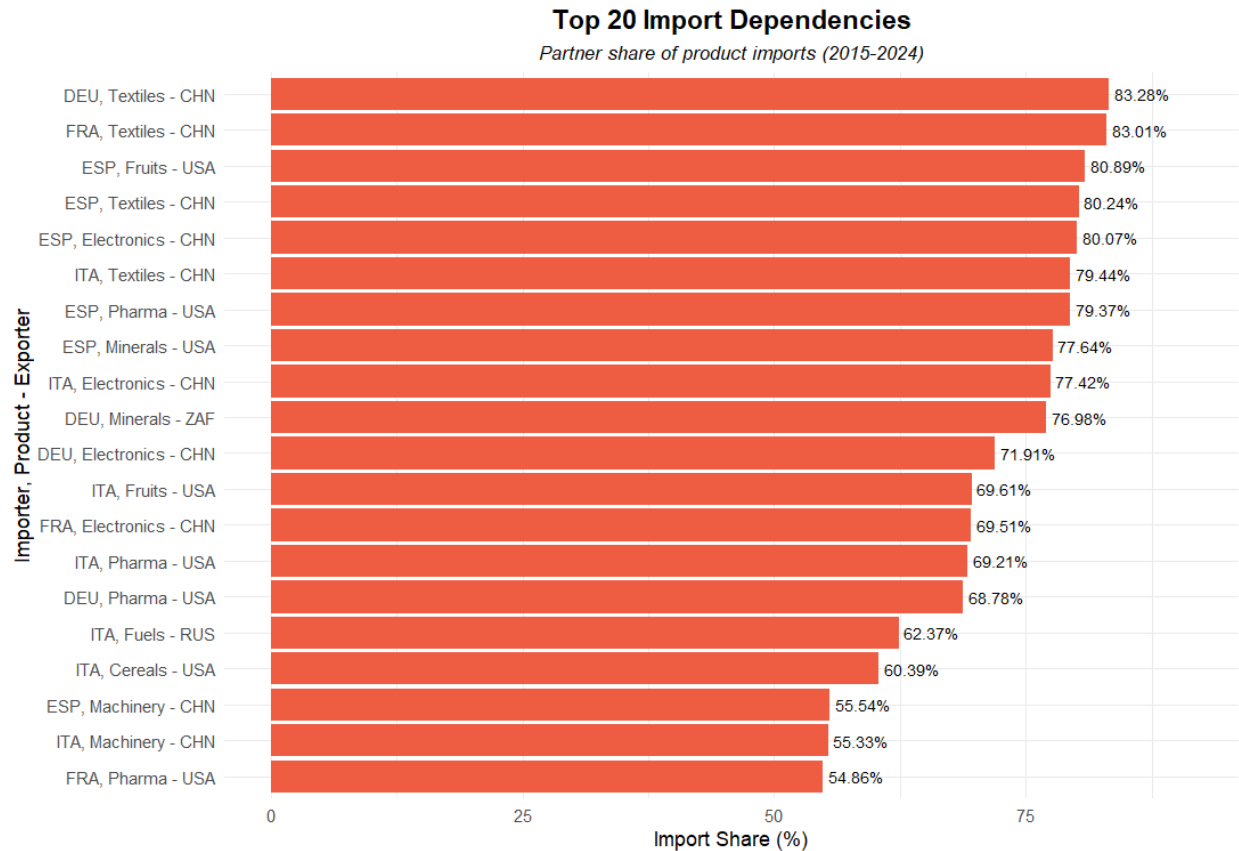


Figure 2: Top 20 Import Dependencies by EU Country and Product (2015-2024)

3.1.2 Country-Level Vulnerabilities

Figure 3 shows which countries have the most concentrated supply chains. Italy has the highest number of vulnerabilities with six dependencies exceeding 60%: Cereals (USA, 60%), Electronics (China, 77%), Fruits (USA, 70%), Fuels (Russia, 62%), Pharma (USA, 69%), and Textiles (China, 79%).

Spain follows with five high-risk dependencies: Electronics (China, 80%), Fruits (USA, 81%), Minerals (USA, 78%), Pharma (USA, 79%), and Textiles (China, 80%).

Germany has four vulnerabilities: Electronics (China, 72%), Minerals (South Africa, 77%), Pharma (USA, 69%), and Textiles (China, 83%).

France shows the least concentration with only two high-risk dependencies: Electronics (China, 70%) and Textiles (China, 83%).

The pattern reveals textiles as a universal vulnerability, all four countries exceed 79% dependence on China, with France and Germany above 83%. Italy and Spain face broader exposure across

both Chinese manufacturing (electronics, textiles) and US agricultural/pharmaceutical imports. Germany and France show more diversified patterns overall.

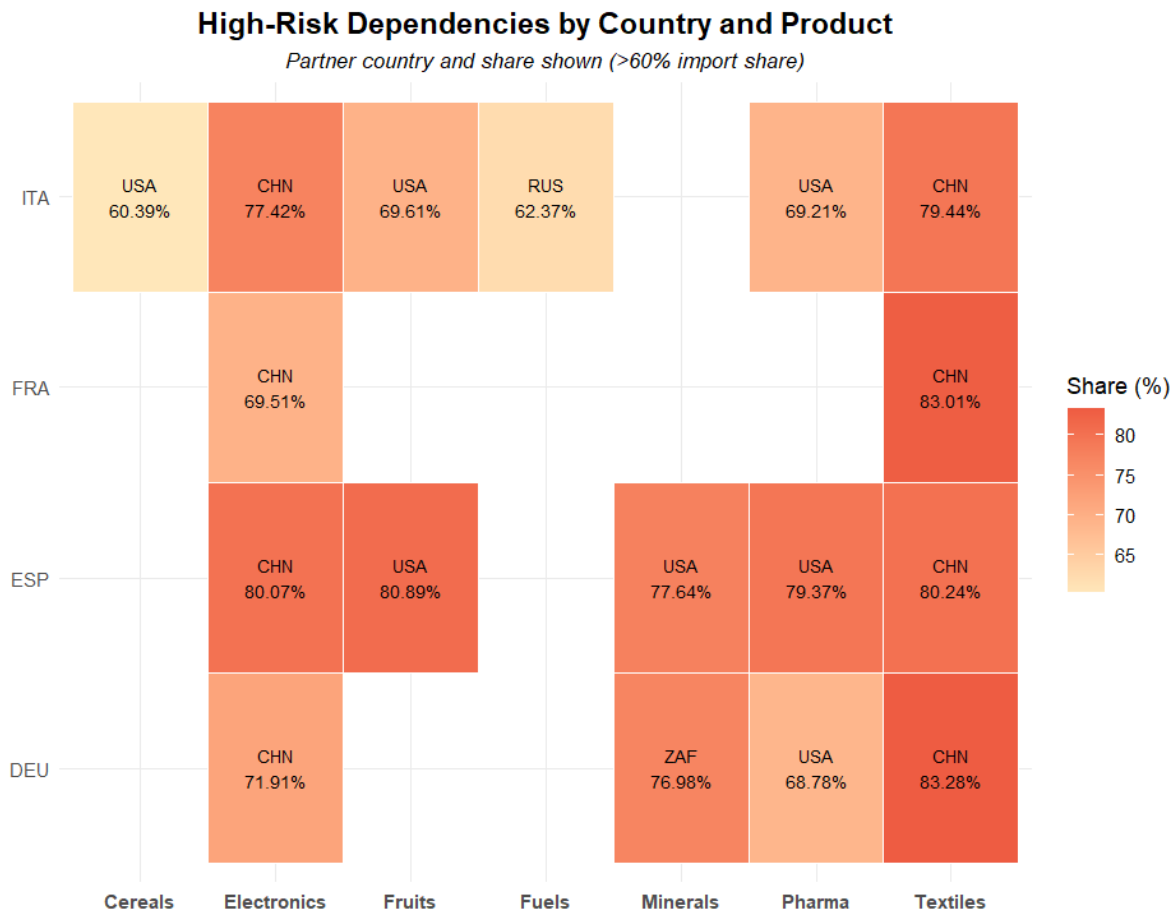


Figure 3: High-Risk Dependencies by Country and Product (>60% Import Share)

3.1.3 Product-Specific Concentration

Figure 4 shows the highest single-partner dependency for each product category. Textiles top the list at 83% (Germany's imports from China). Fruits follow at 81% (Spain from USA) and Electronics at 80% (Spain from China).

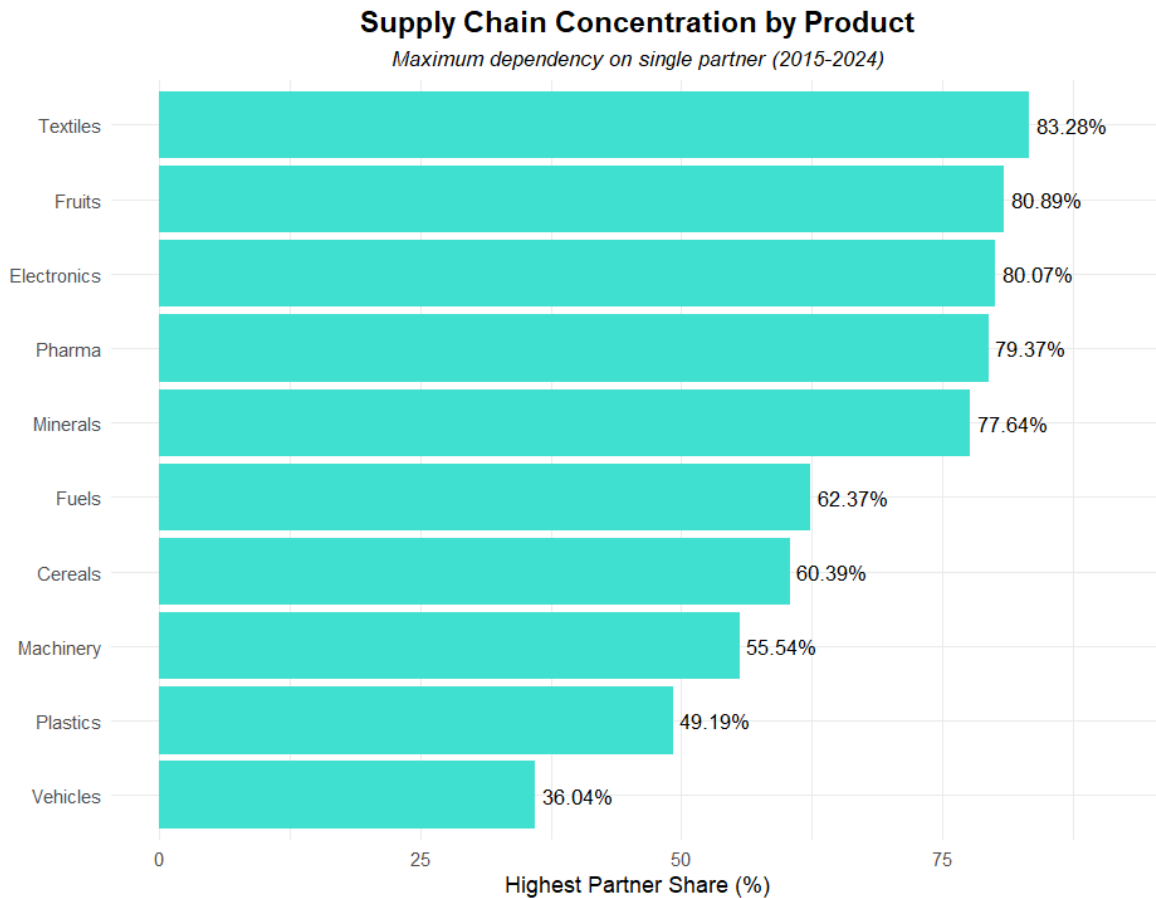


Figure 4: Maximum Supply Chain Concentration by Product

Table 1 shows the complete breakdown. The highest concentration cases are spread across countries: Spain appears six times, Italy three times, and Germany once. China is one of the dominant partners for four products (Textiles, Electronics, Machinery, Plastics), the USA also leads in a tie with four (Fruits, Pharma, Minerals, Cereals). Russia (Fuels) and UK (Vehicles) each appear once.

EU Country	Product	Main Partner	Share (%)
DEU	Textiles	CHN	83.28
ESP	Fruits	USA	80.89
ESP	Electronics	CHN	80.07
ESP	Pharma	USA	79.37
ESP	Minerals	USA	77.64
ITA	Fuels	RUS	62.37
ITA	Cereals	USA	60.39
ESP	Machinery	CHN	55.54
ESP	Plastics	CHN	49.19
ITA	Vehicles	GBR	36.04

Table 1: Highest Import Dependency for Each Product

Three products show relatively low concentrations: Vehicles (36%), Plastics (49%), and Machinery (56%), suggesting more diversified sourcing across EU countries. In contrast, Textiles, Fruits, Electronics, Pharma, and Minerals all exceed 77% in their most concentrated cases.

3.2 Brexit Impact Analysis

3.2.1 Trade Volume Trends

Figure 5 shows EU-UK trade volumes from 2015 to 2024. Both exports and imports dropped sharply in 2020 when Brexit took effect. EU exports to the UK fell from \$90.7 billion (2019) to \$74.1 billion (2020), a drop of 18%. EU imports from the UK declined from \$47.9 billion to \$41.1 billion, a drop of 14%.

Trade partially recovered after 2020 but never returned to pre-Brexit levels. By 2024, exports stabilized around \$90.3 billion and imports around \$45 billion, both still below 2019 values.

Two important notes: First, the 2020 drop reflects both Brexit (January 2020) and COVID-19 (March 2020 onwards), making it impossible to isolate Brexit's specific impact. Second, the partial recovery from 2021-2022 coincided with post-COVID economic reopening, so not all that recovery can be attributed to trade adjustment.

What's clear is that EU-UK trade contracted in 2020 and has not fully recovered four years later.

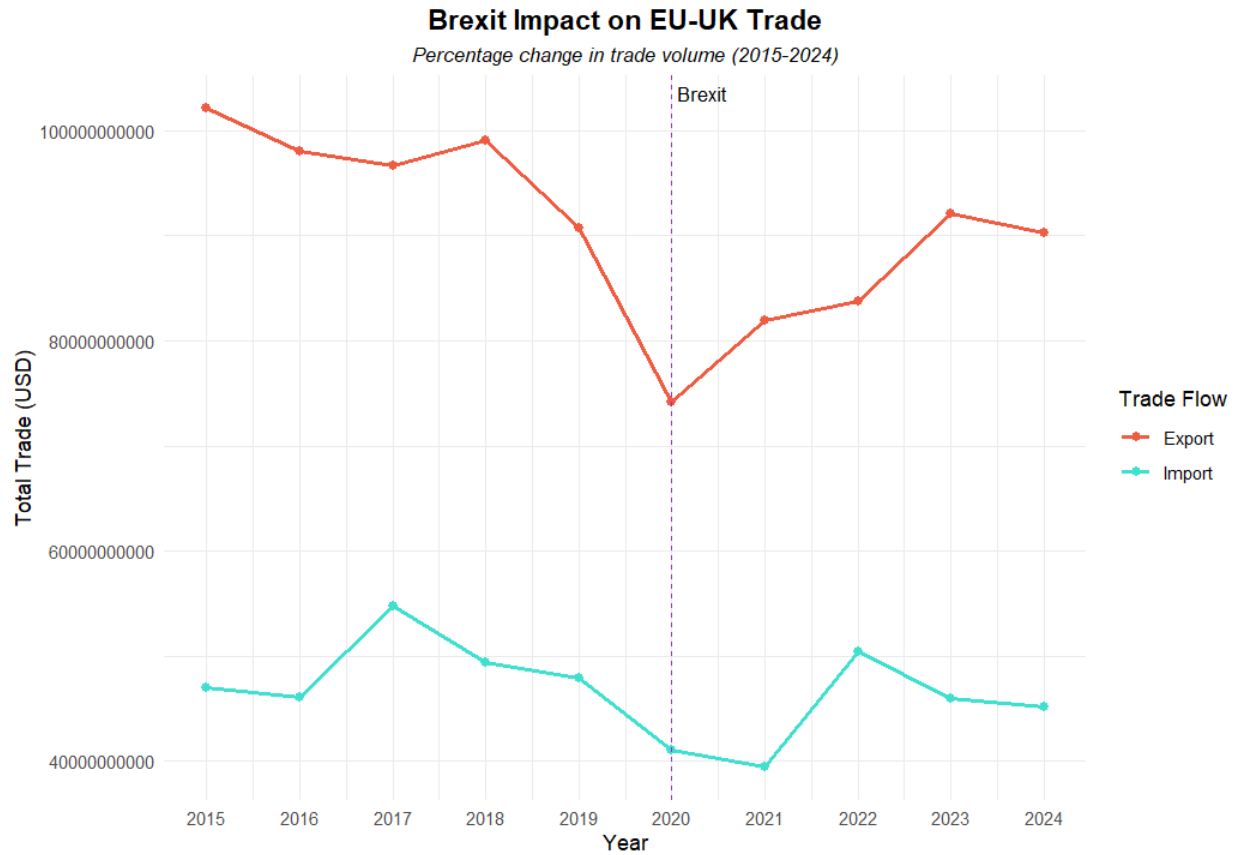


Figure 5: EU-UK Trade Volume Trends (2015-2024)

3.2.2 Partner Substitution Patterns

Table 2 shows how EU countries shifted their import sources for Electronics and Fruits between 2019 and 2023. For products where UK imports declined, the table reveals which partners gained market share.

Product	Partner	Percentage Change (2019-2023)
Electronics	NGA	168.86
Electronics	IND	163.37
Electronics	EGY	77.31
Electronics	CHN	56.03
Electronics	JPN	-0.61
Electronics	USA	-4.67
Electronics	ZAF	-6.88
Electronics	GBR	-28.59
Electronics	RUS	-82.84
Fruits	NGA	136.12
Fruits	EGY	117.39
Fruits	ZAF	36.23
Fruits	RUS	34.56
Fruits	JPN	28.89
Fruits	CHN	-20.11
Fruits	USA	-24.07
Fruits	IND	-50.91
Fruits	GBR	-88.03

Table 2: Import Changes by Partner for Electronics and Fruits (2019-2023)

Electronics: UK imports dropped by 29%, Russia fell even more sharply (-83%), likely due to Ukraine war sanctions. Meanwhile, Nigeria (+169%), India (+163%), Egypt (+77%), and China (+56%) all gained significant share. This suggests EU countries diversified electronics sourcing away from the UK primarily toward Asian and African suppliers.

Fruits: UK fruit imports collapsed by 88%, the steepest drop across both products. Several partners gained: Nigeria (+136%), Egypt (+117%), South Africa (+36%), Russia (+35%), and Japan (+29%). However, the USA (-24%), China (-20%), and India (-51%) all declined.

Key limitation: These patterns only cover the ten analyzed partners. If EU countries shifted to untracked partners like Turkey, Morocco, or Latin American countries, those substitution patterns won't appear in this data.

3.2.3 Product-Level Impacts

Figure 6 shows which products experienced the largest UK import decline between 2019 and 2023. Fruits saw the steepest drop at 88%, followed by Minerals (46%) and Electronics (29%). Plastics (-16%), Pharma (-15%), Vehicles (-8%), and Textiles (-6%) showed smaller declines. The wide variation suggests Brexit's impact was highly product-specific rather than uniform across all categories.

The pattern reveals that agricultural products (fruits) and raw materials (minerals) were hit hardest, while manufactured goods like textiles and vehicles showed relatively modest declines.

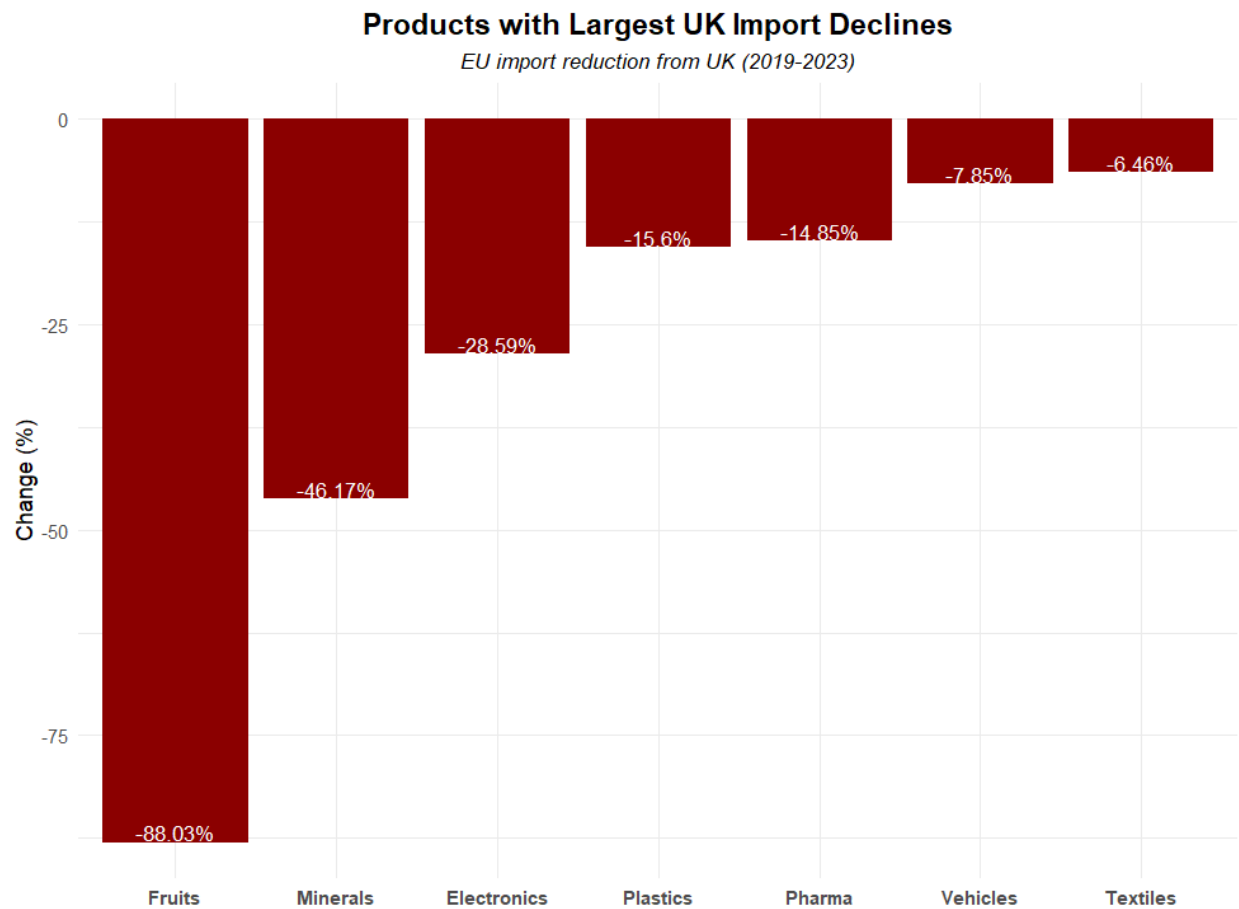


Figure 6: Products with Largest UK Import Declines (2019-2023)

4. Discussion

4.1 Key Findings

This analysis reveals serious trade concentration risks across EU economies. All four countries show extreme dependence on China for textiles (79-83%) and electronics (69-80%). The USA dominates pharmaceutical imports for all countries except France. Italy faces the most vulnerabilities overall with six high-risk dependencies, while France shows the least with only two.

Brexit's impact on EU-UK trade is clear but impossible to isolate from COVID-19. Trade volumes dropped 18% (exports) and 14% (imports) in 2020 and haven't fully recovered. Fruits and minerals saw the steepest UK import declines (88% and 46%), while textiles and vehicles were barely affected. EU countries didn't simply replace UK imports with existing major partners, instead, countries like Nigeria, India, and Egypt gained significant share in electronics and fruits.

4.2 Policy Implications

Textiles represent a universal vulnerability requiring immediate attention. Relying on a single country for over 80% of textile imports creates massive disruption risk. Policymakers should incentivize diversification through trade agreements with alternative suppliers or domestic production support.

Pharmaceutical dependence on the USA is equally concerning. Health security requires multiple supply sources. The EU should develop strategic reserves and strengthen pharmaceutical manufacturing capacity within member states.

Italy and Spain need targeted diversification strategies. Their broad exposure across multiple product-partner combinations makes them particularly vulnerable to any single disruption event.

4.3 Limitations

This analysis captures only ten partners out of hundreds of trading relationships. Major economies like Canada, Brazil, and Turkey are excluded, potentially missing significant diversification that occurred outside the analyzed subset.

The data can't separate Brexit from COVID-19 or Ukraine war effects. The 2020 trade collapse and subsequent patterns reflect multiple simultaneous shocks, not Brexit alone.

Trade values don't reveal resilience capacity. Countries with identical import shares might have vastly different vulnerability depending on stockpiles, alternative suppliers, or domestic production capacity, none of which appears in this data.

5. Conclusion

EU trade dependencies are concentrated and risky. China dominates manufacturing, the USA dominates pharmaceuticals, and most EU countries lack diversified supply chains for critical products. Brexit reduced EU-UK trade significantly, but attribution is impossible due to overlapping shocks.

The path forward requires deliberate diversification policies, strategic reserves for critical products, and stronger intra-EU supply chains. Without action, these dependencies will continue to pose economic and security risks.

References

- United Nations. (2024). *UN Comtrade Database*. Retrieved from <https://comtradeplus.un.org/>